Now, $(ace)^{\frac{1}{p}} = (b^p d^{2p} f^{3p})^{\frac{1}{p}}$

$= (bd^2 f^3)^{\frac{p}{p}}$

$= bd^2f^3$

8. Option (c) is correct.

Given that $-\sqrt{2}$ and $\sqrt{3}$ are roots of the given equation. Therefore, $\sqrt{2}$ and $-\sqrt{3}$ are also roots of the given equation.

$\therefore x^4 + a_3x^3 + a_2x^2 + a_1x + a_0$

$= (x+\sqrt{2})(x-\sqrt{2})(x-\sqrt{3})(x+\sqrt{3})$

$= (x^2 - 2)(x^2 - 3)$

$= x^4 - 5x^2 + 6$

$\therefore a_3 = 0, a_2 = -5, a_1 = 0$ and $a_0 = 6$

9. Option (c) is correct.

Let $z_1 = x_1 + iy_1$ and $z_2 = x_2 + iy_2$

Now, $\left|\frac{z_1 + z_2}{z_1 - z_2}\right| = 1$

$\Rightarrow \quad |z_1 + z_2| = |z_1 - z_2|$

$\Rightarrow |(x_1 + x_2) + i(y_1 + y_2)|$

$= |(x_1 - x_2) + i(y_1 - y_2)|$

$\Rightarrow (x_1 + x_2)^2 + (y_1 + y_2)^2 = (x_1 - x_2)^2 + (y_1 - y_2)^2$

$\Rightarrow x_1^2 + x_2^2 + 2x_1x_2 + y_1^2 + y_2^2 + 2y_1y_2$

$= x_1^2 + x_2^2 - 2x_1x_2 + y_1^2 + y_2^2 - 2y_1y_2$

$\Rightarrow \quad x_1x_2 + y_1y_2 = 0 \qquad$ ...(i)

$$\frac{z_1}{z_2} = \frac{x_1 + iy_1}{x_2 + iy_2} \times \frac{x_2 - iy_2}{x_2 - iy_2}$$

$$= \frac{(x_1x_2 + y_1y_2) + i(y_1x_2 - x_1y_2)}{x_2^2 + y_2^2}$$

$$= 0 + \frac{i(y_1x_2 - x_1y_2)}{x_2^2 + y_2^2}$$

$\therefore \quad \text{Re}\left(\frac{z_1}{z_2}\right) = 0$

Now, $\text{Re}\left(\frac{z_1}{z_2}\right) + 1 = 1$

10. Option (b) is correct.

$\because \quad 26! = n8^k = n2^{3k}$

Maximum power of 2

$$= \left[\frac{26}{2}\right] + \left[\frac{26}{4}\right] + \left[\frac{26}{8}\right] + \left[\frac{26}{16}\right]$$

$= 13 + 6 + 3 + 1 = 23$

But power of 2 is $3k$

$\therefore$ Maximum value of $3k = 21$

$\Rightarrow \quad k = 7$

11. Option (b) is correct.

(1) $\because \quad adj(AB) = (adj\ B)(adj\ A)$

$\therefore$ Statement 1 is wrong.

(2) $\because \quad AB \neq BA$

$\therefore \quad adj(AB) \neq adj(BA)$

$\therefore$ Statement 2 is wrong.

(3) $(AB)\ adj\ (AB) - |AB|\mathrm{I}_n$

$\Rightarrow |AB|\ \mathrm{I}_n - |AB|\ \mathrm{I}_n \qquad [\because A.\ adj\ A = |A|\mathrm{I}]$

= Null matrix

$\therefore$ Statement 3 is correct.

12. Option (d) is correct.

By properties statement 1 is correct.

We know that If $A^n = A$ then A is identity matrix.

$\therefore$ Statements 2 and 3 are correct.

So, all statements are correct.

13. Option (b) is correct.

Even digit = 0, 2, 4, 6, 8.

Choice $\rightarrow$ 4 5 5 5

$\therefore$ Total required number = $4 \times 5 \times 5 \times 5 = 500$

14. Option (b) is correct.

$(z - 100)^3 + 1000 = 0$

$\Rightarrow \quad (z - 100)^3 = (-10)^3$

$\therefore \quad z - 100 = -10\ (w, w^2, 1)$

$\therefore \quad z - 100 = -10$

$\Rightarrow \quad z = 90$

$z - 100 = -10w$

$\Rightarrow \quad z = 100 - 10w$

$z - 100 = -10w^2$

$\Rightarrow \quad z = 100 - 10w^2$

15. Option (d) is correct.

$(1 + i)^4 + (1 - i)^4 = [(1 + i)^2]^2 + [(1 - i)^2]^2$

$= (1 - 1 + 2i)^2 + (1 - 1 - 2i)^2$

$= (2i)^2 + (-2i)^2$

$= 4i^2 + 4i^2$

$= -4 - 4 = -8$

16. Option (a) is correct.

We know that all diagonal elements of skew symmetric matrix is zero then their sum is also zero.

So, statements 1 and 2 are correct.

We know that If $AA^T = \mathrm{I}$ then

A is called orthogonal matrix.

But $\quad AA^T = A(-A) = -A^2 \quad [\because A^T = -A]$

So, Statement 3 is wrong.